

Data Collection and Preprocessing Phase

Date	15/08/2026
Team ID	xxxxxx
Project Title	Indian's agriculture crop production analysis
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset	Missing values in numerical columns (Production, Yield, Area).	High	Apply median imputation for skewed numerical distributions or drop rows where target values are null.
Dataset	Inconsistent categorical values (e.g., spelling differences in Crop or State_Name, leading/trailing whitespace).	Moderate	Standardize strings using <code>.str.strip().str.title()</code> and map known spelling variations using a cleaning dictionary.
Dataset	Extreme outliers in crop yield numbers due to reporting errors.	Moderate	Detect outliers via IQR (Interquartile Range) method and clip or remove erroneous extremes based on historical yield thresholds.
Dataset	Incorrect data types (e.g.,	Low	Convert columns to

Data Source	Data Quality Issue	Severity	Resolution Plan
	numeric production figures stored as object/string types).		appropriate data types using <code>pd.to_numeric()</code> with error coercion to float.